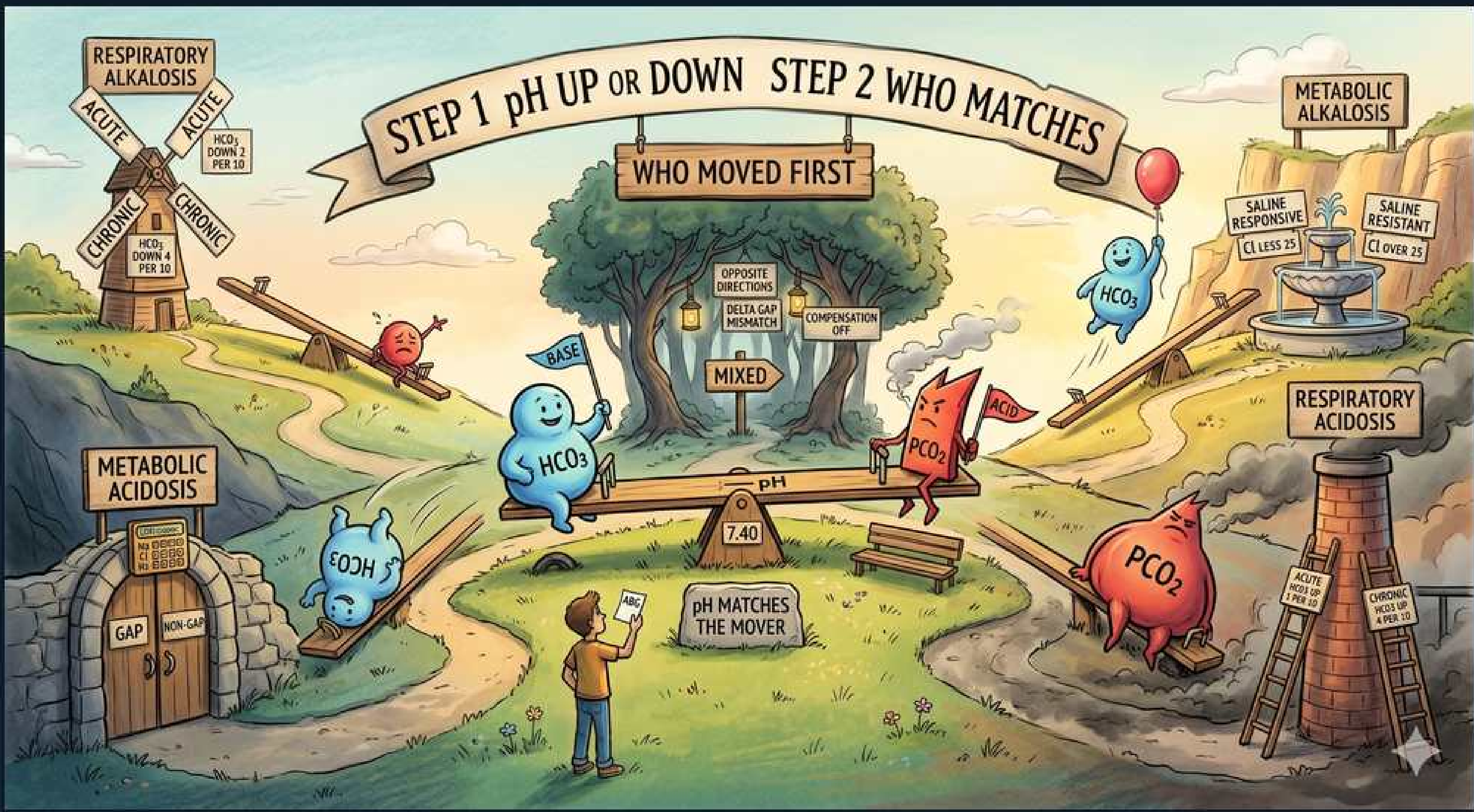
Renal Reference — Acid-Base Approach (Step 1 & 2 Map)



1. Step 1 — pH up or down

WHAT YOU SEE

Banner: STEP 1 — pH UP or DOWN

WHAT IT ENCODES

ALWAYS start by reading the arterial pH: normal is 7.35-7.45 (mean 7.40). pH <7.35 = ACIDEMIA, pH >7.45 = ALKALEMIA; the value tells you the dominant primary process even when compensation has shifted other values. The Henderson-Hasselbalch relationship underlies this: $pH = 6.1 + \log([HCO_3]/(0.03 \times pCO_2))$, so pH is governed by the RATIO of bicarbonate (kidney/metabolic) to dissolved CO_2 (lungs/respiratory).

BOARD TRAP

WRONG to open the analysis by looking at the bicarbonate or pCO_2 first and naming the disorder from there — e.g., seeing a low HCO_3 and calling it 'metabolic acidosis' when the pH is actually alkalemic (compensated metabolic alkalosis or respiratory alkalosis). The discriminator: pH defines the PRIMARY direction; an isolated bicarb or pCO_2 can be compensatory, so reading them before pH leads you to the wrong primary diagnosis.

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2. Step 2 — who matches

WHAT YOU SEE

Banner: STEP 2 — WHO MATCHES / WHO MOVED FIRST

WHAT IT ENCODES

Once you know the pH direction, decide which value — HCO_3 or pCO_2 — explains it: the one whose change moves pH in the SAME direction as the abnormality is the PRIMARY disorder, and the other is COMPENSATION. Recall the directional rules: HCO_3 ↓ or pCO_2 ↑ lowers pH (acidosis); HCO_3 ↑ or pCO_2 ↓ raises pH (alkalosis). Then quantify compensation with the appropriate formula (e.g., Winter's for metabolic acidosis) to detect a hidden second disorder.

BOARD TRAP

WRONG to label any low bicarbonate as a 'metabolic acidosis' regardless of pH. The discriminator: in a respiratory ALKALOSIS the bicarbonate also falls (renal compensation), so a low HCO_3 with a HIGH pH is compensation, not a primary metabolic acidosis — match the mover to the pH direction before assigning the primary disorder.

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3. pH balance beam (7.40)

WHAT YOU SEE

Central seesaw labeled pH with 7.40 fulcrum, HCO_3 figure on one side, PCO_2 figure on the other

WHAT IT ENCODES

Conceptualize pH as a seesaw balancing BASE (HCO_3, controlled slowly by the kidney) against ACID (pCO_2, controlled rapidly by the lungs), pivoting at 7.40. Tilting toward more bicarbonate or less CO_2 raises pH (alkalemia); tilting toward less bicarbonate or more CO_2 lowers pH (acidemia). This base/acid ratio is exactly the Henderson-Hasselbalch numerator/denominator — normal ratio $HCO_3:pCO_2 \approx 24:40$ keeps pH at 7.40.

BOARD TRAP

WRONG to assume the two arms move independently or that a 'normal' pH means no disorder. The discriminator: a normal pH can mask a fully compensated disorder OR opposing mixed disorders (e.g., metabolic acidosis + metabolic alkalosis) — the seesaw can balance even when both ends are abnormal, so always inspect HCO_3 and pCO_2 individually even when pH looks fine.

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4. HCO3 character = metabolic

WHAT YOU SEE

Blue HCO_3 figure waving a 'BASE' flag

WHAT IT ENCODES

Bicarbonate (HCO_3, normal ~22-26 mEq/L) is the METABOLIC/renal arm of acid-base balance — the kidney regulates it slowly (hours to days) by reabsorbing filtered HCO_3 and generating new HCO_3 via ammoniagenesis and titratable acid excretion. A primary FALL in HCO_3 (with matching low pH) = metabolic acidosis; a primary RISE in HCO_3 (with high pH) = metabolic alkalosis. Because renal compensation is slow, respiratory (pCO_2) compensation for a metabolic problem begins within minutes via altered ventilation.

BOARD TRAP

WRONG to treat a low serum bicarbonate on a basic metabolic panel as automatically a metabolic acidosis without a blood gas. The discriminator: the BMP 'CO_2'/total bicarbonate can be low as COMPENSATION for a respiratory alkalosis, so confirm with pH on an ABG — if HCO_3 is the primary mover it will track the pH direction; if it's compensation it won't.

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5. PCO2 character = respiratory

WHAT YOU SEE

Red PCO_2 figure

WHAT IT ENCODES

pCO_2 (normal ~35-45 mmHg, mean 40) is the RESPIRATORY arm and behaves as an ACID gas: the lungs adjust it rapidly (seconds to minutes) by changing minute ventilation. A primary RISE in pCO_2 (hypoventilation, low pH) = respiratory acidosis; a primary FALL in pCO_2 (hyperventilation, high pH) = respiratory alkalosis. Respiratory compensation for metabolic disorders is FAST, whereas metabolic (renal) compensation for respiratory disorders is SLOW, which is why respiratory disorders are split into acute vs chronic.

BOARD TRAP

WRONG to expect full renal compensation immediately in a respiratory disorder. The discriminator: renal (metabolic) compensation takes 3-5 DAYS to develop, so an ACUTE respiratory acidosis shows only a small HCO_3 rise (~1 mEq/L per 10 mmHg pCO_2) versus a CHRONIC one (~3.5 mEq/L per 10) — using the acute expectation in a chronic CO_2 retainer falsely suggests a second disorder.

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6. pH matches the mover

WHAT YOU SEE

Sign: 'pH MATCHES THE MOVER'

WHAT IT ENCODES

The core rule for naming the primary disorder: the value (HCO_3 or pCO_2) whose abnormality moves pH in the SAME direction as the observed pH change is the primary disorder; the opposite-moving value is compensation. Examples: low pH + low HCO_3 → metabolic acidosis (HCO_3 matches); low pH + high pCO_2 → respiratory acidosis (pCO_2 matches); high pH + high HCO_3 → metabolic alkalosis; high pH + low pCO_2 → respiratory alkalosis. Compensation NEVER fully normalizes the pH or over-corrects past 7.40.

BOARD TRAP

WRONG to think compensation can push pH past normal to the opposite side. The discriminator: if the pH has 'overshot' to the other side of 7.40, that is NOT compensation but a SECOND primary disorder — physiologic compensation only partially corrects pH and never crosses the 7.40 fulcrum.

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7. Metabolic acidosis corner

WHAT YOU SEE

Stone gate labeled METABOLIC ACIDOSIS with GAP door (HCO3 down)

WHAT IT ENCODES

Metabolic acidosis = low pH, low HCO3, with compensatory low pCO2 (hyperventilation). The first branch is the ANION GAP (Na − [Cl + HCO3], normal ~8-12, correct for albumin: add ~2.5 per 1 g/dL below 4). HIGH-gap causes (HAGMA) = MUDPILES: Methanol, Uremia, DKA/ketoacidosis, Propylene glycol, Iron/Isoniazid, Lactic acidosis, Ethylene glycol, Salicylates. NORMAL-gap causes (NAGMA, hyperchloremic) = diarrhea (GI HCO3 loss) and renal tubular acidoses — use the URINE anion gap (positive in RTA, negative with diarrhea) to separate them.

BOARD TRAP

WRONG to skip the anion-gap step or to forget to correct it for hypoalbuminemia. The discriminator: a critically ill, hypoalbuminemic patient can have a 'normal-looking' gap that is actually elevated once corrected (add ~2.5 to the gap per 1 g/dL albumin below 4.0) — missing this hides an underlying high-gap (e.g., lactic) acidosis.

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8. Metabolic alkalosis corner

WHAT YOU SEE

Fountain labeled METABOLIC ALKALOSIS — SALINE RESPONSIVE vs SALINE RESISTANT

WHAT IT ENCODES

Metabolic alkalosis = high pH, high HCO3, with compensatory high pCO2 (hypoventilation, blunted by hypoxic drive so pCO2 rarely exceeds ~55). Classify by URINE CHLORIDE: SALINE-RESPONSIVE (urine Cl <20, volume-depleted) — vomiting/NG suction, diuretics, post-hypercapnia — corrects with normal saline; SALINE-RESISTANT (urine Cl >20) — primary hyperaldosteronism, Cushing, Bartter/Gitelman, severe hypokalemia, exogenous mineralocorticoid — does NOT respond to saline. Maintenance of alkalosis requires ongoing volume/Cl/K depletion that impairs renal HCO3 excretion.

BOARD TRAP

WRONG to use URINE SODIUM to assess volume status in metabolic alkalosis. The discriminator: in alkalosis the kidney excretes HCO3 dragging Na with it, so urine Na can be HIGH despite true volume depletion — URINE CHLORIDE is the reliable marker (low = saline-responsive/volume-deplete; high = saline-resistant/mineralocorticoid-driven).

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9. Respiratory alkalosis corner

WHAT YOU SEE

Windmill labeled RESPIRATORY ALKALOSIS — ACUTE vs CHRONIC (HCO3 down per...)

WHAT IT ENCODES

Respiratory alkalosis = high pH, primary low pCO2 (hyperventilation), with compensatory low HCO3. Compensation: ACUTE HCO3 falls ~2 mEq/L per 10 mmHg drop in pCO2; CHRONIC HCO3 falls ~4-5 per 10 (renal, over days). Causes: anxiety/pain, hypoxemia, PULMONARY EMBOLISM, sepsis (early), pregnancy/progesterone, high altitude, hepatic failure, and EARLY salicylate toxicity (direct respiratory-center stimulation).

BOARD TRAP

WRONG to read isolated early-phase salicylate toxicity as a simple respiratory alkalosis and miss the classic MIXED picture. The discriminator: aspirin overdose produces BOTH a primary respiratory alkalosis (CNS stimulation) AND a primary high-anion-gap metabolic acidosis (lactate/ketones/salicylate) — the tip-off is a near-normal pH with low pCO2 AND low HCO3 that don't fit a single compensation formula.

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10. Respiratory acidosis corner

WHAT YOU SEE

Ladder/sign labeled RESPIRATORY ACIDOSIS — ACUTE vs CHRONIC

WHAT IT ENCODES

Respiratory acidosis = low pH, primary high pCO2 (hypoventilation/alveolar hypoventilation), with compensatory high HCO3. Compensation: ACUTE HCO3 rises ~1 mEq/L per 10 mmHg rise in pCO2; CHRONIC HCO3 rises ~3.5-4 per 10 (renal, 3-5 days). Causes: CNS depression (OPIOIDS, sedatives), neuromuscular disease (Guillain-Barré, myasthenia, ALS), chest-wall/severe obesity (OHS), and airway disease — chronic COPD ('blue bloater') is the prototype chronic CO2 retainer.

BOARD TRAP

WRONG to give high-flow oxygen to titrate a chronic CO2 retainer to a normal SpO2 without watching the pCO2. The discriminator: in chronic hypercapnic COPD, aggressive O2 can worsen hypercapnia (loss of hypoxic drive, V/Q changes, Haldane effect) — target SpO2 ~88-92%; also distinguish acute-on-chronic respiratory acidosis (sudden pH drop) from compensated chronic (near-normal pH with high HCO3).

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11. Mixed disorder check

WHAT YOU SEE

Center sign: MIXED — 'opposite directions', 'delta gap mismatch', 'compensation off'

WHAT IT ENCODES

Suspect a MIXED acid-base disorder when: (1) HCO3 and pCO2 move in OPPOSITE/non-physiologic directions (e.g., HCO3 down AND pCO2 up = combined metabolic + respiratory acidosis), (2) the delta-delta is off, or (3) compensation falls OUTSIDE the predicted formula range. Use Winter's formula for metabolic acidosis (expected pCO2 = 1.5 × HCO3 + 8 ± 2) — if measured pCO2 is HIGHER than predicted there's a concurrent respiratory acidosis; if LOWER, a concurrent respiratory alkalosis.

BOARD TRAP

WRONG to assume a single disorder once you've named the primary one and to stop. The discriminator: always verify that compensation matches the predicted formula (Winter's, etc.) — compensation OUTSIDE the expected range means a SECOND primary disorder is present; e.g., a DKA patient who is also vomiting can have metabolic acidosis + metabolic alkalosis hidden together.

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12. Delta gap mismatch

WHAT YOU SEE

Listed clue: 'DELTA GAP MISMATCH'

WHAT IT ENCODES

The delta-delta ratio = ΔAnion Gap (measured AG − 12) ÷ ΔHCO3 (24 − measured HCO3) unmasks additional metabolic disorders coexisting with a high-gap acidosis. Ratio ~1-2 = pure HAGMA; ratio <1 (gap rose less than bicarbonate fell) = concurrent NAGMA (e.g., HAGMA + diarrhea); ratio >2 (bicarbonate higher than the gap rise predicts) = concurrent metabolic ALKALOSIS (e.g., DKA + vomiting). It corrects for the fact that each added acid should drop HCO3 roughly 1:1 with the gap rise.

BOARD TRAP

WRONG to be reassured by a 'normal' serum bicarbonate when the anion gap is elevated. The discriminator: a normal or even high HCO3 with a HIGH anion gap signals a mixed high-gap metabolic acidosis PLUS a metabolic alkalosis (delta-delta >2) — the alkalosis has propped the bicarbonate back up, so always compute the delta-delta rather than trusting the bicarbonate alone.
